

Bahir Dar institute of Technology

Faculty of Mechanical and Industrial Engineering

Mechanical Engineering 3rd year Fluid Mechanics Worksheet

(For assignment Q7-17 Submission date; November 10, 2025)

1. Clearly define lagrangian and Eulerian Approaches in analysis of fluid flow?
2. Explain briefly the type of fluid flow?
3. State Newton's second law of fluid Mechanics and the classification of fluids based on this law with examples?
4. A cylindrical tank of methanol has a mass of 60 kg and a volume of 75 L. Determine the methanol's weight, density, specific weight and specific gravity. Take the gravitational acceleration to be 9.81 m/s^2 .
5. If the pressure in container A in Fig. 2.5 is 150 kPa, compute the pressure in container B.

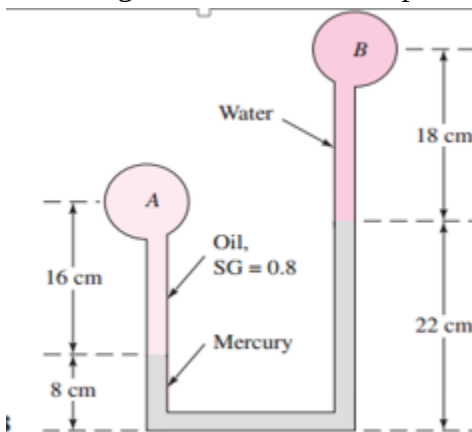


Figure 2.5

6. For the inverted manometer of Fig. 2.6, all fluids are at 20 C. If $p_B - p_A = 97 \text{ kPa}$, what must the height H be in cm?
7. In Fig. 2.7 both the tank and the tube are open to the atmosphere. If $L = 2.13 \text{ m}$, what is the angle of tilt θ of the tube?

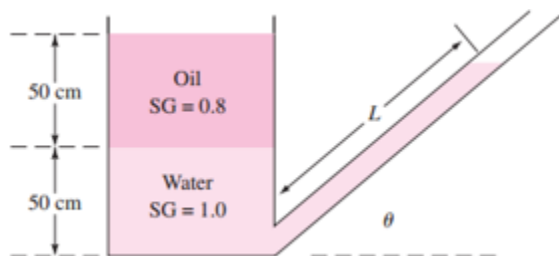


Figure 2.7

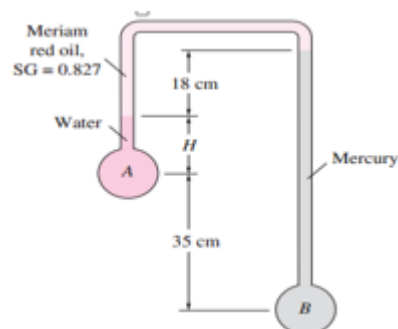


Figure 2.6

8. A gasoline line is connected to a pressure gage through a double-U manometer, as shown in Fig. 2.8. If the reading of the pressure gage is 260 kPa, determine the gage pressure of the gasoline line.
9. Water flows upward in a pipe slanted at 30° , as in Fig. 2.9. The mercury manometer reads $h=12$ cm. Both fluids are at 20°C . What is the pressure difference p_1-p_2 in the pipe?

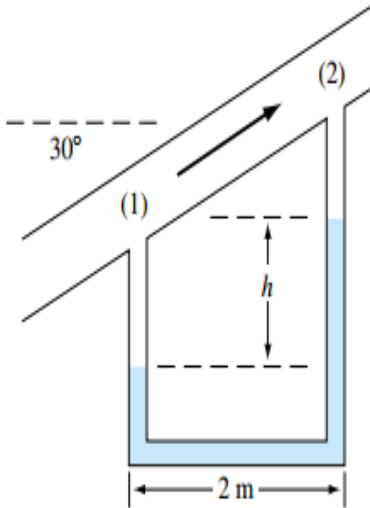


Figure 2.9

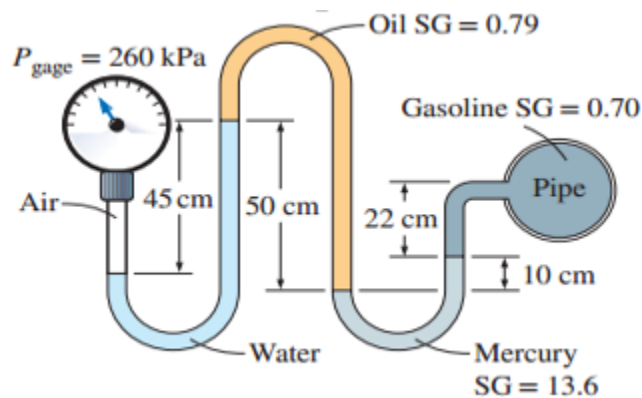


Figure 2.8

10. A 3-m-high, 5-m-wide rectangular gate is hinged at the top edge at A and is restrained by a fixed ridge at B as shown in figure 2.10. Determine the hydrostatic force exerted on the gate by the 5-m-high water and the location of the pressure center.
11. Panel ABC in the slanted side of a water tank is an isosceles triangle with the vertex at A and the base BC 2 m, as in Fig. 2.11 below. Find the water force on the panel and its line of action.

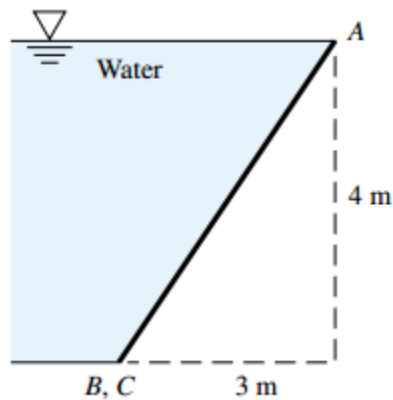


Figure 2.11

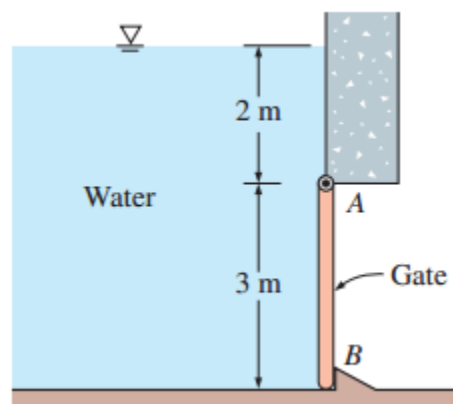


Figure 2.10

12. The dam in Fig.2.12 is a quarter circle 50 m wide into the paper. Determine the horizontal and vertical components of the hydrostatic force against the dam and the point CP where the resultant strikes the dam.

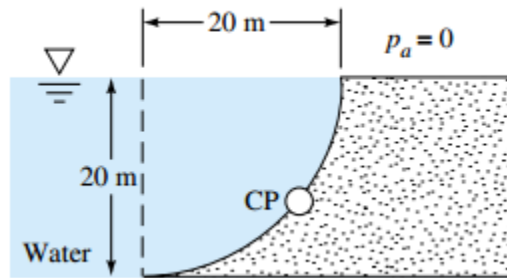


Figure 2.12

13. The concrete (specific weight = 150 lb/ft³) seawall of Fig. P2.13 has a curved surface and restrains seawater at a depth of 24 ft. The trace of the surface is a parabola as illustrated. Determine the moment of the fluid force (per unit length) with respect to an axis through the toe (point A)
14. The quarter circle gate BC in Fig. P2.14 is hinged at C. Find the horizontal force P required to hold the gate stationary. Neglect the weight of the gate

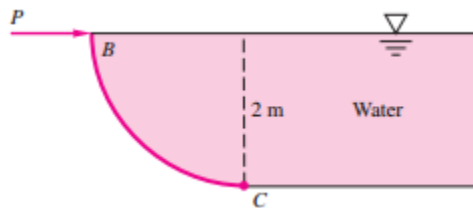


Figure 2.14

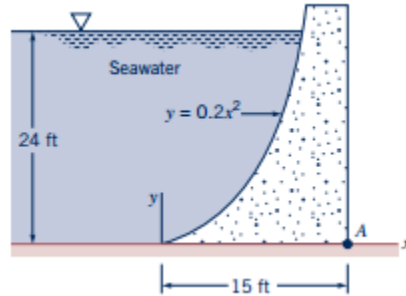


figure 2.13

15. An iceberg can be idealized as a cube of side length L, as in Fig. P2.15. If seawater is denoted by $S = 1.0$, then glacier ice (which forms icebergs) has $S = 0.88$. Determine if this “cubic” iceberg is stable for the position shown in Fig. 2.13.

16. The tank of water in Fig. 2.16 accelerates uniformly by freely rolling down a 30° incline. If the wheels are frictionless, what is the angle θ ? Can you explain this interesting result?

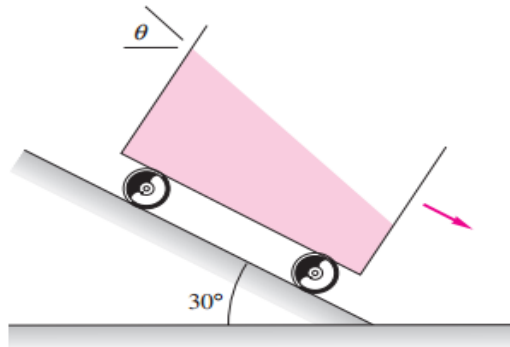


Figure 2.16

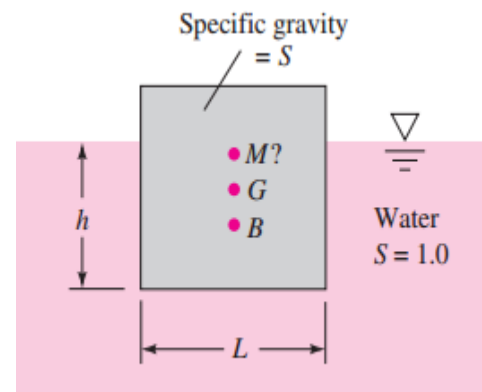


Figure 2.15

17. A 16-cm-diameter open cylinder 27 cm high is full of water. Compute the rigid-body rotation rate about its central axis, in r/min, (a) for which one-third of the water will spill out and (b) for which the bottom will be barely exposed.